

# 21-701: Discrete Mathematics

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# Chapter 1

## Graphs and Colorings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about coloring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a coloring with only two colors. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colors are no longer enough takes real work. Turning the question around—how large must a complete graph be before every coloring of its edges is forced to produce something monochromatic—gives us the Ramsey numbers. The chapter then leaves coloring behind and puts three more questions to hypergraphs: whether one vertex can be chosen out of each edge, all of them different, which is Hall's marriage theorem; how many edges a hypergraph can have once no edge is allowed to sit inside another, which is Sperner's; and how many it can have once no two edges are allowed to miss each other altogether, which is Erdős-Ko-Rado.

### Notation.

- $[k]$  will denote  $\{1, \dots, k\}$ .
- $\sim$  will denote the adjacency relation on graph vertices.
- $K_n$  will denote the complete graph on  $n$  vertices.
- $\mathcal{H}$  will denote the set of hyper-edges of a hypergraph  $H$ , and likewise for any other letter; calligraphic letters will denote families of sets more generally.

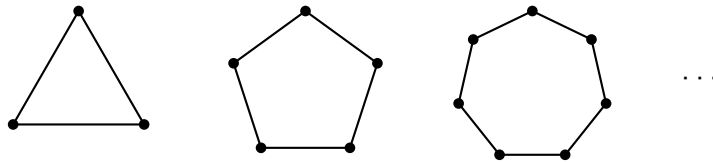
## 1.1 Colorings

**Definition 1.1.1** (Coloring). Given a graph  $G = (V, E)$ , a  $k$ -coloring of  $G$  is a map  $c : V \rightarrow [k]$  such that  $u \sim v \implies c(u) \neq c(v)$ .

### 1.1.1 2-Colorings of Graphs

Let's start with a motivating question: which graphs can be 2-colored?

Any odd cycle graph can't be 2-colored. That is, any graph that looks like



Let's study when odd cycles exist.

**Lemma 1.1.2.** *If  $G$  has an odd closed walk, then  $G$  has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbor, return).

*Proof.* Suppose  $G$  has an odd closed walk. Let  $W$  be such a walk of minimal length. Write  $W = (x_1, \dots, x_i, \dots, x_j, \dots, x_\ell)$ , with  $x_\ell = x_1$ . Suppose  $W$  is not a cycle. Then there must be some  $x_i, x_j$  such that  $x_i = x_j$  with  $1 \leq i < j \leq \ell - 1$ , ie, there must be some repetition beyond the closing one. Then,  $(x_1, \dots, x_i, x_{j+1}, \dots, x_\ell)$  and  $(x_i, x_{i+1}, \dots, x_j)$  are both shorter walks than  $W$ . Moreover, one of them must be of odd length. This contradicts minimality of the length of  $W$ . So  $W$  must be a cycle. □

Here's a cool result.

**Theorem 1.1.3.** *A graph  $G = (V, E)$  can be 2-colored if and only if  $G$  has no odd cycle.*

*Proof.*

( $\implies$ ) We've already seen that odd cycles can't be 2-colored. So if  $G$  can be 2-colored, it can't have an odd cycle. We've essentially seen this from the examples above.

( $\impliedby$ ) We may assume  $G$  is connected, since otherwise we can color each component separately.

Fix some  $v_0 \in V$ . Define  $c : V \rightarrow \{1, 2\}$  by

$$c(u) = \begin{cases} 2 & \text{if } \text{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \text{dist}(v_0, u) \text{ is even} \end{cases}$$

where  $\text{dist}(\cdot, \cdot)$  denotes the shortest number of edges from one vertex to another.

We claim that this  $c$  is a valid 2-coloring.

Indeed, suppose it is not. Then, there must be some  $u, v \in V$  such that  $u \sim v$  and  $c(u) = c(v)$ . Now let's consider the two possible cases on the color.

Suppose  $c(u) = c(v) = 1$  for adjacent  $u, v$ . Then, there is a path of even length between  $v$  and  $v_0$  and another path of even length between  $u$  and  $v_0$ . Concatenating the path from  $v$  to  $v_0$  with the one from  $v_0$  to  $u$ , and closing up with the edge  $uv$ , we get a closed walk of length  $\text{even} + \text{even} + 1$ . This is odd, so by Lemma 1.1.2,  $G$  has an odd cycle.

The case  $c(u) = c(v) = 2$  is identical: the two paths are now of odd length, so the closed walk they form with the edge  $uv$  has length  $\text{odd} + \text{odd} + 1$ , which is odd again.

Either way,  $G$  has an odd cycle, contrary to hypothesis. So  $c$  must be a valid 2-coloring.

□

### 1.1.2 Colorings of Hypergraphs

Coloring is not really a question about graphs. All the definition above asks of  $G$  is that we know which pairs of vertices may not share a color, and nothing stops us forbidding larger sets instead.

**Definition 1.1.4** (Hypergraph). Given a vertex set  $V$ , a **hypergraph** is a pair  $H = (V, \mathcal{H})$ , where  $\mathcal{H}$  is a collection of subsets of  $V$ . The sets in  $\mathcal{H}$  are called **hyper-edges**.

**Definition 1.1.5** (Uniformity). A hypergraph  $H$  is  $k$ -uniform if  $|f| = k$  for all  $f \in \mathcal{H}$ .

The following is thus obvious.

**Proposition 1.1.6.** A graph is a 2-uniform hypergraph.

**Definition 1.1.7.** A proper coloring of a hypergraph  $H$  is an assignment  $c : V \rightarrow [r]$  such that  $\forall f \in \mathcal{H}, |c(f)| \geq 2$ .

We say “no edge is monochromatic”.

We can ask ourselves the following question: “Which 3-uniform hypergraphs are 2-colorable?”

For each  $k$ , define  $m(k)$  to be the smallest  $m$  such that  $\exists$  some  $k$ -uniform hypergraph with  $m$  edges that is not 2-colorable. We can show that  $m(2) = 3$ . But what is  $m(3)$ ? That is, how many 3-edges do I need to prevent 2-colorability?

Let  $H$  be a 3-uniform hypergraph that's not 2-colorable. Assume, further, that  $H$  has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where  $H$  has exactly 6 vertices. Indeed, if  $H$  has vertices  $x, y$  that are not in any common edge, then I can define  $H'$  by “merging”  $x$  and  $y$ . So  $H'$  has one fewer vertex than  $H$ . This  $H'$  cannot be 2-colorable: if it were, then that coloring would extend to  $H$  by using the color of the merged vertex at both  $x$  and  $y$ , which would make  $H$  2-colorable. And when  $H$  has more than 6 vertices, such a pair always exists.

An edge has 3 vertices, so it covers the  $\binom{3}{2} = 3$  pairs among them, and  $H$  has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if  $H$  has  $n \geq 7$  vertices, there are  $\binom{n}{2} \geq \binom{7}{2} = 21$  pairs to cover. So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that  $H$  has exactly 6 edges and 6 vertices and is not 2-colorable.

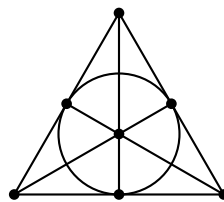
There is no loss in the vertex count in the other direction either: a hypergraph with fewer than 6 vertices can be handed as many extra vertices as it likes, and vertices lying in no edge change nothing about which colorings are proper.

Now count colorings. Among all the ways of coloring 6 vertices with two colors, consider only the balanced ones, which split  $V$  into two sets of three; there are  $\frac{1}{2}\binom{6}{3} = 10$  of these. Such a coloring fails exactly when some edge is monochromatic, and an edge of  $H$  has three vertices, so it is monochromatic under a balanced coloring precisely when it is one of the two color classes. Each edge  $f$  therefore rules out exactly one of the ten balanced colorings, namely the one splitting  $V$  into  $f$  and  $V \setminus f$ . Six edges rule out at most six of the ten, so at least four survive, and any one of them is a proper 2-coloring of  $H$ . This contradicts our assumption, so no such  $H$  exists and

$$m(3) \geq 7$$

That bound is attained, and the hypergraph attaining it is a familiar object.

**Example 1.1.8** (The Fano Plane). The **Fano plane** is the 3-uniform hypergraph on the seven vertices below, whose seven edges—by convention, its **lines**—are the three sides of the triangle, the three medians, and the circle through the three edge midpoints.



Every vertex lies on exactly three lines, any two vertices lie on exactly one line together, and any two lines meet in exactly one vertex. That last property is what does the work below.

**Proposition 1.1.9.** *The Fano plane is not 2-colorable. Hence  $m(3) = 7$ .*

*Proof.* Of the  $\binom{7}{4} = 35$  sets of four vertices, exactly 28 contain a line: a line together with any one of the four vertices off it gives such a set, which is  $7 \cdot 4 = 28$  sets in all, and no two of them coincide because four vertices cannot contain two lines—two lines meet in a vertex, so between

them they cover five. The seven sets left over are the complements of the seven lines.

Now suppose we had a proper 2-coloring, with color classes  $A$  and  $B$ , and say  $|A| \leq 3$ . If  $|A| = 3$ , then  $A$  is not a line, since  $A$  is monochromatic; so  $B = V \setminus A$  is a set of four vertices that is not the complement of a line, and hence contains one. If  $|A| \leq 2$ , then the vertices outside  $B$  lie on at most  $3 + 3 - 1 = 5$  lines between them, so at least two lines avoid them both and lie inside  $B$ . Either way some line is monochromatic, which is a contradiction.

So the Fano plane is a 3-uniform hypergraph with seven edges that is not 2-colorable, giving  $m(3) \leq 7$ .  $\square$

### 1.1.3 Bounds on $m(k)$

Pinning down  $m(3)$  took a page of case analysis, and doing the same for  $m(4)$  is out of the question. What we can do for every  $k$  at once is bound it, and the lower bound falls out of an idea that will keep coming back: color at random, and show that a bad coloring is improbable.

**Theorem 1.1.10.**  $m(k) \geq 2^{k-1}$ . In other words, any  $k$ -uniform hypergraph with  $< 2^{k-1}$  edges is 2-colorable.

*Proof.* Given such a hypergraph, randomly color the vertices red and blue, with equal probability ( $1/2$ ) for each. For each edge  $f$ , let  $\mathcal{E}_f$  be the event that  $f$  is monochromatic. Then,

$$\Pr(\mathcal{E}_f) = \frac{2}{2^k}$$

Taking a simple union bound,

$$\Pr\left(\bigcup_{f \in \mathcal{H}} \mathcal{E}_f\right) \leq \frac{|\mathcal{H}|}{2^{k-1}} < 1$$

where we interpret  $\bigcup_{f \in \mathcal{H}} \mathcal{E}_f$  as the event that there is a monochromatic edge. As the probability is less than 1, there must be at least one coloring with no monochromatic edge.  $\square$

Note that the above argument can also be written combinatorially instead of probabilistically. In this instance, the probabilistic language essentially conveys the same information as combinatorial language would. But this may not always be true, so it's worth being careful about this.

In the other direction, the truth is only a factor of about  $k^2$  away.

**Exercise 1.1.11.** Prove that  $m(k) = O(k^2 \cdot 2^k)$

### 1.1.4 Maker-Breaker Games

Theorem 1.1.10 says that a hypergraph with few enough edges has a proper 2-coloring somewhere among all the colorings of its vertices. It turns out that the very same threshold survives a far more demanding requirement: that one of two players, claiming vertices one at a time, be able to stop the other from ever owning a whole edge. Winning that does not by itself produce a proper 2-coloring, since the winner's own class may contain a monochromatic edge. It does force one to exist all the same: a breaker who can win going second can also win going first, by holding a spare vertex in reserve, and setting those two strategies against each other leaves neither class containing an edge.

Here's a fun fact: a "maker-breaker" game is a game where the "maker" wins if they are able to achieve an objective and the "breaker" wins if they are able to stop the maker from winning. So if you set up a "maker-breaker" version of tic-tac-toe, where only the crosses can win by making three in a row and the noughts can only win by stopping the crosses, the crosses will always win (assuming the crosses play first). This is, of course, different from tic-tac-toe, which, as our own studies from when we were a kid have shown, can always be drawn.

The threshold at which breaker can hold out is exactly the one we have just been looking at.

**Theorem 1.1.12** (Erdős-Selfridge). *If  $H$  is a  $k$ -uniform hypergraph with  $m$  edges, and  $m < 2^{k-1}$ , then breaker wins (if breaker goes second).*

In the language of hypergraphs, the game is played on the vertices of  $H$ : the two players alternately claim unclaimed vertices, maker wins by claiming every vertex of some edge, and breaker wins if play ends with no edge wholly maker's.

*Proof.* Let  $\Delta$  be the maximum degree of  $H$ , that is,

$$\Delta = \max_{v \in V} |\{f \in \mathcal{H} \mid v \in f\}|$$

We'll prove that if  $m + \Delta < 2^k$ , then breaker will win. Then, since  $\Delta \leq m$ , it will follow that if  $m < 2^{k-1}$ , then  $m + \Delta \leq 2m < 2^k$ .

At the start of move  $i$ , we will conceptualize the danger of edge  $f \in \mathcal{H}$  by

$$\phi_i(f) = \begin{cases} 2^{\text{number of maker moves in } f} & \text{if there are no breaker moves in } f \\ 0 & \text{if there is a breaker move in } f \end{cases}$$

$$\Phi_i = \sum_{f \in \mathcal{H}} \phi_i(f)$$

Consider the strategy where, once maker has played  $y$  on move  $i$ , breaker plays the unclaimed vertex carrying the most danger, that is, the  $x$  maximizing

$$\sum_{f \ni x} \phi(f)$$

with  $\phi$  measured after maker's move. Two things move  $\Phi$  on move  $i$ . Maker's play doubles the danger of every edge through  $y$  that breaker has not already killed, so it raises  $\Phi$  by  $a_i = \sum_{f \ni y} \phi_i(f)$ ; breaker's play sends the danger of every surviving edge through  $x$  to 0, so it lowers  $\Phi$  by the quantity  $b_i$  that breaker has just maximized. Hence

$$\Phi_{i+1} = \Phi_i + a_i - b_i$$

The point of breaker's rule is that  $a_i \leq b_{i-1}$  for every  $i \geq 2$ . Indeed,  $y$  is unclaimed right up until maker takes it on move  $i$ , so it was one of the vertices breaker was choosing between on move  $i - 1$ , and breaker chose one carrying at least as much danger as it. Between then and the start of move  $i$  nothing happens except breaker's own play, which only ever sends dangers to 0, so the danger through  $y$  at the start of move  $i$  is no greater than it was when breaker passed  $y$  over.

Summing  $\Phi_{i+1} = \Phi_i + a_i - b_i$  over the first  $n$  moves and canceling each  $a_i$  against  $b_{i-1}$ , everything past the first two terms disappears and we are left with

$$\Phi_{n+1} \leq \Phi_1 + a_1$$

Stopping the same sum one term earlier gives the same bound halfway through a move, after maker has played and before breaker has replied. Now  $\Phi_1 = m$ , since at the start every edge contains a move of neither player and so contributes  $2^0 = 1$ , and  $a_1$  is the number of edges through maker's

opening vertex, which is at most  $\Delta$ . So the danger never exceeds  $m + \Delta$  at any point in the game.

But if maker ever completes an edge  $f$ , then at that moment  $f$  contains all  $k$  of its vertices from maker and none from breaker, so  $\phi(f) = 2^k$  and hence  $\Phi \geq 2^k$ . Since  $m + \Delta < 2^k$ , this never happens, and breaker wins.  $\square$

Let's consider higher-dimensional maker-breaker tic-tac-toe.

**Example 1.1.13.** Turns out that  $8 \times 8 \times 8$  maker-breaker tic-tac-toe is breaker-winnable: we can show that  $m = \frac{10^3 - 8^3}{2} = 244$  and  $\Delta = 7$ , meaning  $m + \Delta = 251 < 2^k = 256$ .

Both numbers come out of the geometry. For  $m$ , pad the board with one extra cell in every direction, giving a  $10 \times 10 \times 10$  cube, and run each winning line one step past each of its two ends. The two cells we land on lie in the shell we just added, and every cell of the shell is the end of exactly one line: a coordinate of the cell equal to 0 tells the line to increase along that axis, one equal to 9 tells it to decrease, and the rest stay fixed. So the lines are in bijection with those pairs of shell cells that arise as the two ends of a line, and since every shell cell is the end of exactly one line there are  $\frac{10^3 - 8^3}{2}$  of them. For  $\Delta$ , a line through a given cell is determined by its direction, and a corner cell has the 3 lines along the axes, the 3 face diagonals and the one long diagonal through it, which is 7; no cell of an  $8 \times 8 \times 8$  board does better. And it is the  $m + \Delta < 2^k$  form of the criterion doing the work here, since 244 is a long way above  $2^{k-1} = 128$ .

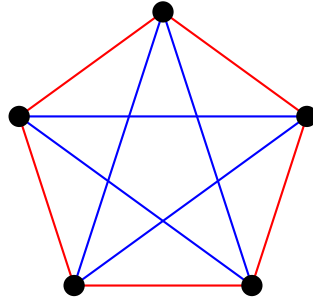
We won't say any more about tic-tac-toe in this course, but it was a fun aside.

## 1.2 Ramsey Numbers

Everything so far has asked when a coloring exists. The opposite question is at least as natural: how large does a structure have to be before every coloring of it is forced to contain something monochromatic? Let's step back into "graph land" from "hypergraph land" and ask it of the complete graphs, coloring edges this time rather than vertices.

**Definition 1.2.1** (Ramsey Number). The  $k$ th Ramsey number, denoted  $R(k)$ , is the smallest  $n \in \mathbb{N}$  such that any 2-coloring of edges of  $K_n$  gives a monochromatic  $K_k$ .

One immediate result we can see is that  $R(3) \geq 6$ , which we can see simply by drawing a picture of  $K_5$  and studying what happens if we move our attention to  $K_6$ .



Each color carries a 5-cycle—the outer pentagon in red, the inner pentagram in blue—and a 5-cycle contains no triangle at all, so neither color gives us a monochromatic  $K_3$ . Nothing forces a monochromatic triangle at  $n = 5$ , and so  $R(3) \geq 6$ . The picture also shows how little room there is to spare: every vertex of  $K_5$  meets exactly two edges of each color, and the moment we pass to  $K_6$  each vertex has five edges to distribute between two colors, so one of the colors is forced to take three of them.

We can actually prove that  $R(3) = 6$ , but this is rather nontrivial.

Other known Ramsey numbers are  $R(2) = 2$  and  $R(4) = 18$ . The rest are unknown, though we do have bounds for them. The best (or close to the best) known bounds are

$$2^{\frac{k}{2}} \leq R(k) \leq 3.8^k$$

## 1.3 Matchings in Bipartite Graphs

Hypergraphs came up in Definition 1.1.4 as the setting in which the coloring question gets hard, and we set them aside when the Ramsey numbers took us back to graphs. This section asks something else of them: given a hypergraph, can we pick one vertex out of each edge, all of them different? Getting to the answer means first noticing that a hypergraph, a 0, 1 matrix and a bipartite graph are three views of the same data.

### 1.3.1 Adjacency Matrices

We can represent the data of a hypergraph  $H$  in a matrix, known as the adjacency matrix.

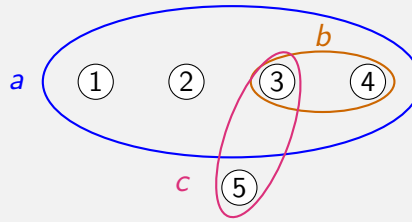
**Definition 1.3.1** (Adjacency Matrix). If  $H$  is a hypergraph with  $n$  vertices and  $m$  hyper-edges, both taken in some fixed order, then the adjacency matrix of  $H$ , denoted  $\text{Adj}(H)$ , is the unique  $n \times m$  matrix  $(a_{ij})$  where

$$a_{ij} = \begin{cases} 1 & \text{if vertex } i \text{ lies in hyper-edge } j \\ 0 & \text{otherwise} \end{cases}$$

Here are some examples of constructing adjacency matrices.

**Example 1.3.2** (Some Adjacency Matrices of Hypergraphs).

1. Consider the hypergraph



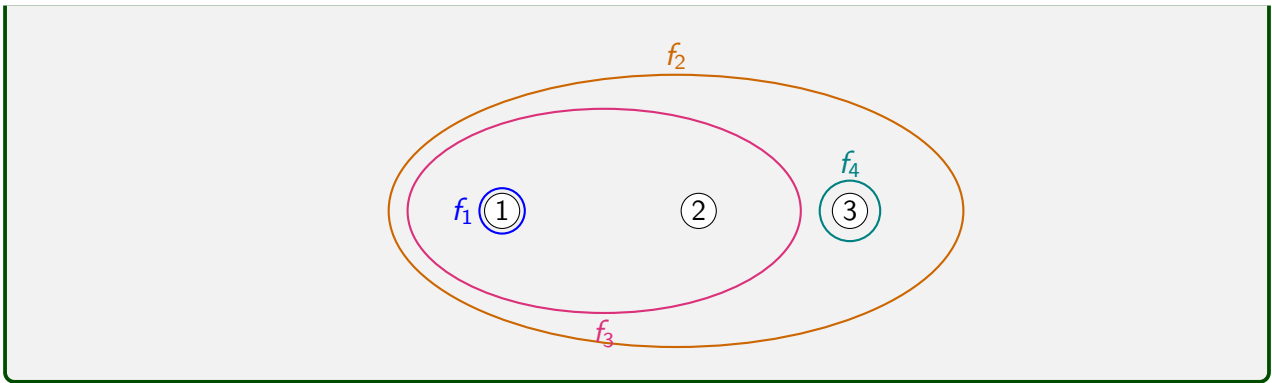
Its adjacency matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

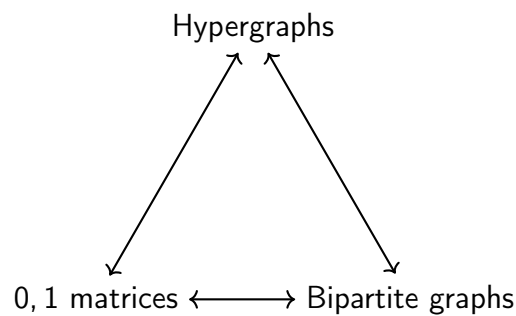
2. Consider the hypergraph

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Reading the columns off one at a time, this is the hypergraph on three vertices with the four hyper-edges  $f_1 = \{1\}$ ,  $f_2 = \{1, 2, 3\}$ ,  $f_3 = \{1, 2\}$  and  $f_4 = \{3\}$ .



It turns out that there is not just a correspondence between hypergraphs and 0,1 adjacency matrices, we can also complete the trifecta with bipartite graphs.



The bipartite graph attached to  $H$  has the vertices of  $H$  on one side and the hyper-edges of  $H$  on the other, with  $v$  joined to  $f$  exactly when  $v \in f$ , which is to say exactly when the adjacency matrix has a 1 in position  $(v, f)$ .

### 1.3.2 Hall's Marriage Theorem

Recall, now, the most important result about bipartite graphs.

**Theorem 1.3.3** (Hall's Marriage Theorem). *Given a bipartite graph with parts  $A$  and  $B$ , with  $A$  finite, there is a complete matching of  $A$  iff  $\forall S \subseteq A$ ,*

$$|\Gamma(S)| \geq |S|$$

where  $\Gamma(S) = \{v \in B \mid \exists u \in S \text{ s.t. } u \sim v\}$ .

*Proof.*

( $\implies$ ) Let  $M$  be a complete matching of  $A$  and let  $S \subseteq A$ . The partners under  $M$  of the elements of  $S$  are  $|S|$  distinct vertices of  $B$ , each adjacent to something in  $S$ , so  $\Gamma(S)$  has at least  $|S|$  elements.

( $\impliedby$ ) We induct on  $|A|$ . If  $|A| = 1$ , then Hall's condition applied to  $A$  itself says that the one vertex of  $A$  has a neighbor, and matching it to one is a complete matching. So suppose  $|A| > 1$ , and let's do some casework.

- Case 1: Assume that in fact,  $\forall S \subsetneq A$  nonempty,  $|\Gamma(S)| \geq |S| + 1$ . Arbitrarily match some pair: pick any  $a \in A$  and any  $b \in \Gamma(\{a\})$ , which is nonempty by hypothesis. After deleting this edge—and with it the vertices  $a$  and  $b$  and every other edge at either of them—the rest satisfies the desired result, since a nonempty  $T \subseteq A \setminus \{a\}$  is a nonempty proper subset of  $A$  and so had at least  $|T| + 1$  neighbors to begin with, of which deleting  $b$  can take away only one. What is left has  $|A| - 1$  vertices on the left, so we are done by induction: it has a complete matching of  $A \setminus \{a\}$ , and adding the edge  $ab$  to that matching completes it to one of  $A$ .
- Case 2: Now assume the opposite, ie, that  $\exists S \subsetneq A$  nonempty such that  $|\Gamma(S)| = |S|$ . We will apply induction again, on  $|A|$  as before.

If we restrict our attention to  $S$ , obviously we have a perfect matching between  $S$  and  $\Gamma(S)$ : every  $T \subseteq S$  has  $\Gamma(T) \subseteq \Gamma(S)$ , so the graph they span satisfies Hall's condition and induction gives a complete matching of  $S$ , which the fact that they're of equal cardinality makes perfect. (There may be more than one, but there is one.)

Define  $A' = A \setminus S$ ,  $B' = B \setminus \Gamma(S)$ . We claim that the bipartite graph induced on  $A'$  and  $B'$  also has a complete matching of  $A'$ , in which case we'd be done (by induction).

Writing  $\Gamma'$  for the neighborhood taken in that smaller graph, we have  $\Gamma'(T) = \Gamma(T) \setminus \Gamma(S)$  for every  $T \subseteq A'$ , and  $\Gamma(S \cup T) = \Gamma(S) \cup \Gamma(T)$  as it does for any two sets, so

$$|\Gamma'(T)| = |\Gamma(S \cup T)| - |\Gamma(S)| = |\Gamma(S \cup T)| - |S| \geq |S \cup T| - |S| = |T|$$

where the inequality is Hall's condition applied to  $S \cup T$ , and the last step is that  $T \subseteq A'$  is disjoint from  $S$ . (If  $\Gamma(T)$  is infinite the display needs no reading at all:  $\Gamma(S)$

is finite, so  $\Gamma'(T)$  is infinite and certainly at least  $|T|$ .) So the smaller graph satisfies Hall's condition, and  $|A'| < |A|$  because  $S$  is nonempty, so induction gives a complete matching of  $A'$  into  $B'$ . That matching touches no vertex of  $S$  and no vertex of  $\Gamma(S)$ , so laying it alongside the matching of  $S$  onto  $\Gamma(S)$  gives a complete matching of  $A$ .

□

Given a hypergraph  $H$ , a family of distinct representatives is a collection of distinct  $x_1, \dots, x_m$  such that  $x_i \in f_i$ , where we have distinct hyper-edges  $f_1, \dots, f_m$  in  $\mathcal{H}$ . Such a collection exists iff for any subcollection  $\mathcal{C}$  of  $f_1, \dots, f_m$ ,

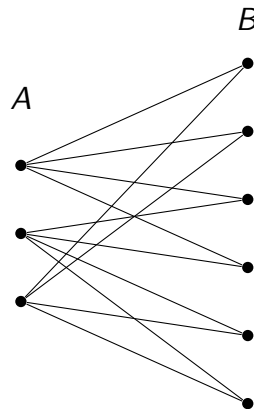
$$\left| \bigcup_{f \in \mathcal{C}} f \right| \geq |\mathcal{C}|$$

In other words, the number of vertices in all the edges in the collection should be at least the number of edges in the collection. This is what Hall's Marriage Theorem tells us about hypergraphs.

Hall's condition asks something of every subset of  $A$ , which is a lot to check. There is a consequence of it that only asks about degrees.

**Theorem 1.3.4.** *If  $G$  is a bipartite graph with parts  $A$  and  $B$ , if all degrees in  $A$  are  $d_1$  and all degrees in  $B$  are  $d_2$ , with  $1 \leq d_2 \leq d_1$ , then there is a complete matching of  $A$ .*

Such a graph is a good deal more rigid than a bipartite graph with Hall's condition alone; here is one with  $d_1 = 4$  and  $d_2 = 2$ .



*Proof.* Given a set  $S \subseteq A$ , there are  $d_1 |S|$  edges coming out of  $S$  and going to the right. Each

vertex of  $\Gamma(S)$  absorbs at most  $d_2$  of them, so

$$|\Gamma(S)| \geq \frac{d_1 |S|}{d_2} \geq |S|$$

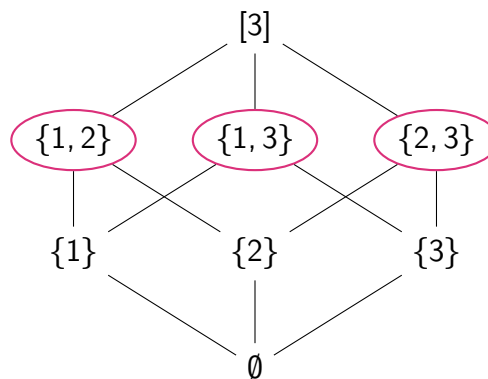
so we can just apply Theorem 1.3.3. □

## 1.4 Sperner Systems

A hypergraph is a vertex set together with a family of subsets of it, and the last question this chapter asks about one is how large that family can be. Unrestricted the question is dull—all  $2^n$  subsets of  $[n]$  form a hypergraph—so we forbid the cheapest way of making a family large, which is to pile sets up inside one another.

**Definition 1.4.1** (Sperner System). A collection of subsets of  $[n]$  such that none is a subset of another is called a Sperner system.

**Example 1.4.2.** The three two-element subsets of  $[3]$ , circled below, form a Sperner system.



The picture is the whole collection of subsets of  $[3]$  ordered by inclusion, arranged in layers by size, with an edge drawn from each set to every set one size up that contains it. Containment between two subsets is then visible as an upward path, so a Sperner system is precisely a collection of vertices no two of which are joined by such a path—and drawing the diagram turns a question about set inclusions into a question about which vertices we are allowed to pick. The diagram as a whole is emphatically not a Sperner system; each individual layer is, since two distinct sets of the same size never contain one another. That gives us two Sperner systems of size 3 here,

the singletons and the two-element subsets, and the top and bottom layers give two more, though useless ones: any system containing  $\emptyset$  or  $[3]$  can contain nothing else.

How large can a Sperner system be?

**Theorem 1.4.3** (Sperner's Theorem). *A Sperner system on  $[n]$  can have at most  $\binom{n}{\lfloor n/2 \rfloor}$  sets.*

An intuitive way of arguing is to find complete matchings of the  $k$ th layer into the  $(k + 1)$ th layer (cf. Theorem 1.3.4), but at some point, this starts to break down. And at what point is that? Before  $k$  exceeds  $n/2$ .

*Proof.* Let  $\mathcal{H}$  be a Sperner system on  $[n]$ . Define

$$\mathcal{F} = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}, \dots, \{1, 2, \dots, n\}\}$$

Choose a random permutation  $\pi \in S_n$ , and define

$$\mathcal{H}_\pi = \{\pi(h) \mid h \in \mathcal{H}\}$$

Define the random variable

$$X = |\{f \in \mathcal{F} \mid f \in \mathcal{H}_\pi\}|$$

Note that we can express

$$X = \sum_{h \in \mathcal{H}} \xi_{h,\pi}$$

where  $\xi_{h,\pi}$  is 1 if  $\pi(h) \in \mathcal{F}$  and 0 otherwise.

Computing the expected value of  $X$ , we get

$$\mathbb{E}(X) = \sum_{h \in \mathcal{H}} \mathbb{E}(\xi_{h,\pi})$$

Let's now compute the expectation of  $\xi_{h,\pi}$ . This is the probability that the permutation  $\pi \in S_n$  sends  $h \in \mathcal{H}$  to some element of  $\mathcal{F}$ . Since you cannot modify sizes, the only possible element in  $\mathcal{F}$  to which  $\pi$  can send  $h$  is precisely the set  $\{1, 2, \dots, |h|\}$ . How many such  $\pi$  can do this? We

have to consider the way  $\pi$  acts on  $h$ , and the way it acts on the other  $n - |h|$  elements of  $[n]$ , because we know it sends  $h$  to  $\{1, 2, \dots, |h|\}$ . There are  $|h|!(n - |h|)!$  such  $\pi$ , so the probability is

$$\frac{|h|!(n - |h|)!}{n!} = \frac{1}{\binom{n}{|h|}}$$

The conclusion now comes from bounding  $\mathbb{E}(X)$  on the other side. Any  $\pi$  is a bijection and so preserves containment, which makes  $\mathcal{H}_\pi$  a Sperner system; and  $\mathcal{F}$  is a chain, any two of its members being comparable. A chain and a Sperner system share at most one set, so  $X \leq 1$  for every  $\pi$ , and hence  $\mathbb{E}(X) \leq 1$ . Since  $\binom{n}{k}$  is largest at  $k = \lfloor n/2 \rfloor$ , this gives

$$1 \geq \mathbb{E}(X) = \sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{|h|}} \geq \sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{\lfloor n/2 \rfloor}} = \frac{|\mathcal{H}|}{\binom{n}{\lfloor n/2 \rfloor}}$$

and rearranging is exactly the bound we wanted. □

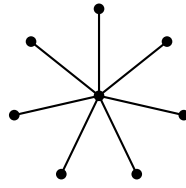
## 1.5 Intersecting Families

Sperner's theorem came out of forbidding one hyper-edge to sit inside another. Forbidding the opposite—a pair of hyper-edges that miss each other entirely—turns out to have a much more interesting answer, and it is the one we spend this section on.

**Definition 1.5.1** (Intersecting). We say that a  $k$ -uniform hypergraph  $H$  is **intersecting** if  $\forall h_1, h_2 \in \mathcal{H}, h_1 \cap h_2 \neq \emptyset$ .

The question is: how big can an intersecting  $k$ -uniform hypergraph on  $n$  vertices be? I.e., how many hyper-edges can it have? Small  $k$  is worth doing by hand, and as long as  $n \geq 2k$  every case points the same way—until the last one, where dropping that assumption is exactly the point.

- $k = 1$ : we can see that the maximum number of edges must be 1.
- $k = 2$ : the maximum number of edges is  $n - 1$ , achieved by the ‘star graph’



- $k = 3$ :  $\binom{n-1}{2}$ , achieved by fixing one vertex that lies in everything and choosing 2 more.
- $k \leq \frac{n}{2}$ :  $\binom{n-1}{k-1}$  by exactly the same strategy.
- $k > \frac{n}{2}$ : we can do better!  $\binom{n}{k}$  works because of something involving pigeons and holes.

The middle of that list is the general answer, and the bottom of it is why  $n \geq 2k$  has to be assumed to get one.

**Theorem 1.5.2** (Erdős-Ko-Rado). *If  $n \geq 2k$ , and  $H$  is an intersecting  $k$ -uniform hypergraph on  $n$  vertices, then the number of hyper-edges of  $H$  is  $\leq \binom{n-1}{k-1}$ . Moreover, if  $n \geq 2k + 1$ , then the (“unique”) extremal example is given by all  $k$ -subsets containing a fixed vertex.*

As a warmup case, let's assume that  $k \mid n$ . Suppose  $H$  is intersecting. Let  $\mathcal{F}$  denote some partition of  $[n]$  into  $n/k$  subsets of size  $k$  (so  $\mathcal{F}$  is fundamentally a collection of subsets).

$$X = |\{h \in \mathcal{H} \mid h = \pi(f) \text{ for some } f \in \mathcal{F}\}|$$

where  $\pi \in S_n$  is a permutation chosen uniformly at random.

Note that  $X \leq 1$ . Let's now compute its expected value.

$$\begin{aligned} \mathbb{E}(X) &= \sum_{h \in \mathcal{H}} \sum_{f \in \mathcal{F}} \Pr(h = \pi(f)) \\ &= |\mathcal{H}| \cdot \frac{n}{k} \cdot \frac{1}{\binom{n}{k}} \end{aligned}$$

So since  $X \leq 1$ , we conclude that

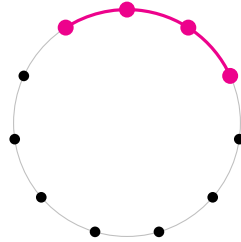
$$|\mathcal{H}| \cdot \frac{n}{k} \cdot \frac{1}{\binom{n}{k}} \leq \mathbb{E}(X) \leq 1$$

Rearranging gives us that

$$|\mathcal{H}| \leq \binom{n}{k} \cdot \frac{k}{n} = \binom{n-1}{k-1}$$

Now let's prove the result in full generality. The random permutation survives, but the partition into  $n/k$  blocks does not, and what replaces it is a cycle.

**Local Notation.** Lay the  $n$  vertices out around a circle. A  $k$ -**interval** of that cycle is a set of  $k$  vertices occupying consecutive positions on it, and  $\mathcal{F}_\pi$  will denote the collection of all of them when the vertices are laid out in the order  $\pi(1), \pi(2), \dots, \pi(n)$ .



There are  $n$  intervals in  $\mathcal{F}_\pi$ , one starting at each vertex, and—this is the whole point of moving to a cycle—very few of them can pairwise intersect.

**Lemma 1.5.3.** *Let  $n \geq 2k$ . Any collection  $\mathcal{C} \subseteq \mathcal{F}_\pi$  of pairwise intersecting  $k$ -intervals has  $|\mathcal{C}| \leq k$ . Moreover, if  $n \geq 2k + 1$  and  $|\mathcal{C}| = k$ , then  $\mathcal{C}$  is the collection of all the  $k$ -intervals containing some one vertex.*

*Proof.* Assume  $\mathcal{C}$  is nonempty and fix some  $I \in \mathcal{C}$ , occupying positions  $a, a+1, \dots, a+k-1$ . For  $i = 1, \dots, k-1$ , let  $A_i$  be the interval ending at position  $a+i-1$  and let  $B_i$  be the one starting at position  $a+i$ . Every interval other than  $I$  that meets  $I$  is one of these  $2k-2$ : an interval meets  $I$  exactly when it starts at one of the  $2k-1$  positions  $a-k+1, \dots, a+k-1$ , and the  $A_i$  take the  $k-1$  of those before  $a$  while the  $B_i$  take the  $k-1$  after it. Now  $A_i$  and  $B_i$  between them occupy the  $2k$  consecutive positions  $a+i-k, \dots, a+i+k-1$ , which are distinct because  $n \geq 2k$ , so  $A_i \cap B_i = \emptyset$  and  $\mathcal{C}$  can contain at most one of the two. Taking  $I$  together with one interval from each of the  $k-1$  pairs gives  $|\mathcal{C}| \leq k$ .

Suppose now that  $n \geq 2k+1$  and  $|\mathcal{C}| = k$ , so that  $\mathcal{C}$  holds exactly one of  $A_i, B_i$  for every  $i$ . It cannot hold both  $A_i$  and  $B_{i+1}$ : those occupy positions  $a+i-k, \dots, a+i-1$  and  $a+i+1, \dots, a+i+k$ , which lie among  $2k+1$  consecutive positions and so are distinct, leaving the two intervals disjoint. So as soon as some  $B_j$  lies in  $\mathcal{C}$  so does every  $B_i$  with  $i < j$ , and hence there is an  $m$  with

$$\mathcal{C} = \{A_{k-1}, \dots, A_{m+1}, I, B_1, \dots, B_m\}$$

Reading off where each of those starts, they are the intervals starting at positions  $a+m+1-k, \dots, a+m$ , which is a run of  $k$  consecutive positions—and the intervals starting there are precisely the ones containing the vertex at position  $a+m$ .  $\square$

The hypothesis  $n \geq 2k+1$  does real work in that second half: on  $C_6$  the three 3-intervals  $\{1, 2, 3\}$ ,  $\{3, 4, 5\}$  and  $\{5, 6, 1\}$  pairwise intersect, and there are  $k = 3$  of them, but no vertex lies in all three.

*Proof of the bound in Theorem 1.5.2.* Choose  $\pi \in S_n$  uniformly at random and set

$$X = |\{h \in \mathcal{H} \mid h \in \mathcal{F}_\pi\}|$$

The intervals being counted here are hyper-edges of  $H$ , so they pairwise intersect, and by Lemma 1.5.3 it is always true that  $X \leq k$ . Now, let's compute the expectation of  $X$ .

$$\mathbb{E}(X) = \sum_{h \in \mathcal{H}} \Pr(h \in \mathcal{F}_\pi)$$

A given  $h$  is an interval of the randomly ordered cycle exactly when  $\pi$  lays its  $k$  elements out consecutively, and we can count the  $\pi$  that do so. There are  $n$  places for the interval to start,  $k!$  ways of arranging  $h$  within it and  $(n-k)!$  ways of arranging the other  $n-k$  vertices, and no  $\pi$  gets counted twice, since an interval determines where it starts. So

$$\Pr(h \in \mathcal{F}_\pi) = \frac{n \cdot k! (n-k)!}{n!} = \frac{n}{\binom{n}{k}}$$

and hence

$$|\mathcal{H}| \cdot \frac{n}{\binom{n}{k}} = \mathbb{E}(X) \leq k$$

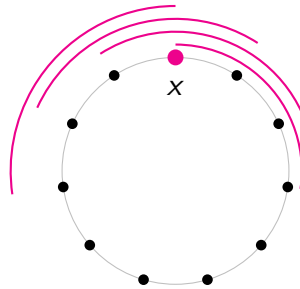
Rearranging,

$$|\mathcal{H}| \leq \frac{k}{n} \binom{n}{k} = \binom{n-1}{k-1}$$

□

That argument has slack in exactly one place, and at  $|\mathcal{H}| = \binom{n-1}{k-1}$  there is none left:  $X$  is then equal to  $k$  for every ordering rather than merely on average. Once  $n \geq 2k + 1$ , so that the second half of Lemma 1.5.3 is available, every ordering of the cycle hands us a run of  $k$  intervals all of which are hyper-edges—and that pins  $\mathcal{H}$  down completely. The following is the “moreover” of Theorem 1.5.2, made precise.

**Theorem 1.5.4.** *Let  $H$  be an intersecting  $k$ -uniform hypergraph on  $n$  vertices. If  $|\mathcal{H}| = \binom{n-1}{k-1}$  and  $n \geq 2k + 1$ , then the extremal example is the obvious one: there is a vertex  $x$  such that  $\mathcal{H}$  consists of precisely the  $k$ -subsets of  $[n]$  that contain  $x$ .*



Whatever order we lay the vertices out in, the picture looks like the one above: the hyper-edges we can see on the cycle are the  $k$  intervals through one distinguished vertex, drawn here as concentric arcs. All the work is in showing that the distinguished vertex does not move when the order changes.

*Proof of Theorem 1.5.4.* Run the argument above again, this time at equality. If  $|\mathcal{H}| = \binom{n-1}{k-1}$ , then

$$\mathbb{E}(X) = \binom{n-1}{k-1} \cdot \frac{n}{\binom{n}{k}} = k$$

and we already know that  $X \leq k$  whatever  $\pi$  is. There are finitely many  $\pi$  and each is equally likely, so a mean of  $k$  under a ceiling of  $k$  leaves nothing to spare:  $X = k$  for every single  $\pi$ . However

we lay the  $n$  vertices out around a cycle, then, exactly  $k$  of that cycle's  $k$ -intervals are hyper-edges of  $H$ , and by Lemma 1.5.3 those  $k$  intervals are the intervals through a single vertex—one vertex and no more, since the outermost two of them share only it. Write  $x_\pi$  for that vertex. For  $k = 1$  an intersecting family is a single hyper-edge and the theorem is immediate, so assume from here that  $k \geq 2$ .

It remains to see that  $x_\pi$  does not depend on  $\pi$ , and one adjacent transposition at a time is enough. Fix an ordering, put  $x = x_\pi$  at position 0 and number the other positions  $1, \dots, n-1$  round the circle, and let  $I_j$  be the interval occupying positions  $j, j+1, \dots, j+k-1$ , indices read modulo  $n$ . The intervals through position  $c$  are  $I_{c-k+1}, \dots, I_c$ , so the hyper-edges of this ordering are exactly  $I_{-(k-1)}, \dots, I_0$ . Transposing the vertices at adjacent positions  $q$  and  $q+1$ , neither of them position 0, changes which vertices occupy  $I_j$  exactly when  $I_j$  holds one of those two positions and not the other—which happens for  $j = q+1$  and for  $j = q-k+1$ , and for no other  $j$ .

So the new ordering's hyper-edges, again the intervals through a single position  $c$ , agree with  $I_{-(k-1)}, \dots, I_0$  outside those two indices. But two runs of  $k$  consecutive indices whose right-hand ends sit  $c$  apart differ in  $2 \min(|c|, k)$  places, where  $|c|$  is the distance from  $c$  to 0 measured around the circle, so  $c \neq 0$  would force  $|c| = 1$ , and the two indices they differ in are then  $\{-(k-1), 1\}$  if  $c = 1$  and  $\{-k, 0\}$  if  $c = -1$ . Either set would have to be  $\{q-k+1, q+1\}$ . Both pairs differ by  $k$ , and matching them in the only order that does not ask for  $-k \equiv k \pmod{n}$ —impossible, as  $n > 2k$ —gives  $q = 0$  in the first case and  $q = -1$  in the second. Both are excluded, so  $c = 0$ : the distinguished vertex is the one at position 0, which the transposition left where it was.

Adjacent transpositions avoiding position 0 generate every arrangement of the other  $n-1$  vertices around the circle, and each one leaves the distinguished vertex at position 0 for the next to start from, so the paragraph above applies afresh along the whole chain. Rotating or reflecting a layout leaves its intervals, and hence  $x_\pi$ , alone—so every  $\pi$  can first be turned until  $x$  sits at position 0, and  $x_\pi = x$  for all of them. Take any  $k$ -subset  $h$  of  $[n]$  containing  $x$ , and lay the vertices out with the elements of  $h$  consecutive: then  $h$  is an interval containing  $x$ , so  $h \in \mathcal{H}$ . There are  $\binom{n-1}{k-1}$  such  $h$ , which is exactly  $|\mathcal{H}|$ , so they are all of it.  $\square$

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